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# Genome-wide analysis of the CCCH zinc finger gene family in pineapple (*Ananas comosus* L.) and their involvement in fruit development and translucency

Zhuanying Yang<sup>1†</sup>, Wenhao Jiang<sup>1†</sup>, Lidan Wang<sup>1</sup> and Dongbo Lin<sup>1\*</sup>

## Abstract

**Background** Pineapple (*Ananas comosus* L.), an important tropical fruit with global significance, encounters ongoing difficulties in maintaining commercial quality due to fruit translucency. However, the molecular mechanisms underlying fruit translucency remain poorly characterized.

**Results** In this study, we explored the CCCH zinc finger (CCCH-ZF) gene family, recognized for its regulatory functions in plant development and stress response, as a potential factor influencing translucency in pineapple. We performed a comprehensive genome-wide analysis, identifying 40 CCCH-ZF genes (*AcC3H1*-*AcC3H40*) categorized into nine phylogenetic groups. Structural analysis revealed conserved CCCH domains and diverse gene architectures, most of which included introns. *Cis*-regulatory element prediction indicated the presence of hormone-responsive motifs (ABA, GA, SA, IAA, MeJA) and growth-related elements, suggesting complex regulatory roles. Expression profiling showed tissue-specific expression patterns, with eight genes predominantly expressed in floral organs. During fruit development, *AcC3H10/14/31* exhibited peak expression at later stages, while *AcC3H2/5* showed a decline. Notably, twelve genes (e.g., *AcC3H7/9/11*) were upregulated during late-stage translucency, whereas four (e.g., *AcC3H1/8*) were downregulated. Subcellular localization suggested nuclear targeting for eight proteins (e.g., *AcC3H4/8*-YFP) and plasma membrane localization for *AcC3H2/16*-YFP, indicating functional diversification. These findings present insights into the structural, evolutionary, and spatial characteristics of pineapple CCCH-ZF genes and highlight their potential roles in fruit development and translucency.

**Conclusion** This study establishes a foundation for future functional analyses to elucidate the molecular mechanisms governing pineapple fruit quality and postharvest physiology.

**Keywords** CCCH zinc finger, Genome-wide analysis, Pineapple, Fruit translucency

<sup>†</sup>Zhuanying Yang and Wenhao Jiang contributed equally to this work.

\*Correspondence:

Dongbo Lin  
dongbolin@gdou.edu.cn

<sup>1</sup>College of Coastal Agricultural Sciences, Guangdong Ocean University, Zhanjiang, Guangdong 524088, PR China



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## Background

As a globally significant tropical fruit crop, pineapple (*Ananas comosus* L.) ranks among the top three most extensively cultivated tropical fruits worldwide and represents the sole commercially cultivated species within the Bromeliaceae family [1]. Pineapple fruit translucency has become a significant and ongoing challenge for the preservation of commercial fruit quality. Translucency is a physiological disorder observed in fruit flesh, characterized by a water-soaked appearance [2]. The pathogenesis of pineapple translucency is a complex process involving various factors, such as changes in gene expression, physiological parameters, endogenous hormone levels, and environmental and cultivation management conditions [2]. However, the molecular regulatory mechanisms underlying fruit translucency in pineapple are not fully understood.

CCCH-type zinc finger proteins are characterized by a distinctive motif comprising three cysteine residues (C) and one histidine (H) residue, and members of this family have been identified across a diverse range of organisms, including yeast, humans and various plant species [3, 4]. The CCCH domain, by coordinating with zinc ions, enables sequence-specific binding to DNA and recognition of RNA, thus playing a crucial role in the regulation of gene expression at the transcriptional or post-transcriptional level [5]. Plant genomes encode large numbers of CCCH zinc-finger proteins across various model plant species. Genomic analyses have revealed the presence of 68 CCCH zinc-finger protein genes in *Arabidopsis thaliana* [6], 67 in *Oryza sativa* [6], 68 in *Zea mays* [7], 80 in *Solanum lycopersicum* [8], and 91 in *Populus trichocarpa* [9]. CCCH zinc-finger protein genes exhibit multifunctional roles and are integral to plant development as well as the response to environmental stresses [10]. *AtC3H17* regulates the processes of vegetative growth, flowering, and seed development in *Arabidopsis* [11]. In rice, the overexpression of *OsTZF1* has been observed to lead to a postponement in seed germination, a reduction in the rate of vegetative growth, and a delay in the process of leaf senescence [12]. *PuC3H35* enhances the drought stress tolerance by regulating proanthocyanidin (PA) biosynthesis and lignin biosynthesis-related genes in the roots of *Populus ussuriensis* [13]. *SIC3H39* negatively modulates cold tolerance in tomato [14]. Moreover, *MaCCCH33*-like 2 actively regulates banana fruit ripening by modulating the genes involved in starch and cell wall degradation [15]. The recent availability of multiple tropic plant genome sequences have provided an opportunity to explore a large set of multigenic families at the genome scale [16]. The CCCH zinc finger gene family has been systematically identified in Banana and Pitaya [17, 18]. However, compared with that, the pineapple CCCH zinc finger gene family has so far been poorly described.

In this study, we addressed the structural, evolutionary, and subcellular localization of the pineapple CCCH zinc finger gene family. Moreover, we investigated the expression profiles of *AcCCCH* genes at various stages of translucency development, with the objective of elucidating their possible roles in fruit translucency. These results will serve as a foundation for future studies to better understand the potential functions of CCCH zinc finger proteins in pineapple plant, especially fruit development and translucency.

## Methods

### Identification of the CCCH zinc finger gene family members in pineapple

CCCH zinc finger protein sequences of *Arabidopsis* and Rice were retrieved from the Plant Transcription Factor Database (<https://planttfdb.gao-lab.org/>) [19]. Members of the CCCH zinc finger gene family were identified by Blastp search in pineapple genome database using *Arabidopsis* and Rice CCCH proteins as query sequences. The hidden Markov model (HMM) for the CCCH domain (PF00642) in the PFAM database (<http://pfam.xfam.org/>) were utilized to search for *AcCCCH* proteins. All candidate genes were further analyzed using Pfam and SMART database (<http://smart.embl-heidelberg.de>) to confirm the presence of conserved CCCH domains.

### Multiple sequence alignment and phylogenetic analysis

To investigate the evolutionary relationships of CCCH proteins in *Arabidopsis thaliana*, *Oryza sativa* and *Ananas comosus*, multiple sequence alignments were performed by ClustalW, and the obtained results were used to construct a phylogenetic tree via the maximum likelihood (ML) method in MEGA 7.0 [20]. The phylogenetic tree was visualized using the Evolview tool (<http://www.evolgenius.info/evolview/>). The CCCH zinc finger protein gene family in *Ananas comosus* was classified on the basis of their evolutionary relationships with the CCCH members in *Arabidopsis*. All of gene ID were listed in Table S3.

### Gene structure, conserved motifs and cis-element analyses of the pineapple CCCH genes

All genomic sequences of pineapple were downloaded from Phytozome database ([https://phytozome-next.jgi.doe.gov/info/Acomosus\\_v3](https://phytozome-next.jgi.doe.gov/info/Acomosus_v3)) [21], and then the structures of the *AcCCCH* genes were identified using the Gene Structure Display Server (<http://gsds.cbi.pku.edu.cn/>). Conserved motifs within the *AcCCCH* proteins were identified using the MEME motif discovery tool (<http://meme-suite.org/index.html>). 2-kb sequences located upstream of the transcription start site for the 40 *AcCCCH* genes were extracted from pineapple genomes using TBtools-II software [22]. Cis-regulatory elements

located within the promoter regions were identified utilizing the PlantCare database (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) and PlantPan3 databases (<http://PlantPAN.its.ncnu.edu.tw>). The obtained data were collated in Table S1, and visualized using TBtools-II software.

#### Physicochemical properties and chromosome location analyses of the pineapple *CCCH* genes

Sequences of the 40 *AcCCCH* protein were analyzed using the online platform ExPASy (<http://web.expasy.org/protparam/>) to calculate their amino acid lengths (AA), molecular weights (kDa) and theoretical pI. The chromosomal positional information for pineapple was obtained from the Pineapple Genomics Database (PGD: <http://pineapple.zhangjisenlab.cn/pineapple/html/index.html>). The 40 *AcCCCH* genes were mapped to their corresponding chromosomes using Mapchart 2.32 software.

#### Subcellular localization of *AcCCCH* proteins

Subcellular localization of the *AcCCCH* proteins was predicted using Cell-PLoc 2.0 (<http://www.csbio.sjtu.edu.cn/bioinf/Cell-PLoc-2/>) [23]. The obtained data were collated in Supplementary Table S1. For localization of the selected *AcCCCH* proteins, the *AcCCCH* coding sequences were inserted into pC29\_35S: YFP vector using homologous recombination technology to generate *AcC3Hs*-YFP plasmids. Primers used in constructing plasmids were listed in Supplementary Table S2. *Agrobacterium tumefaciens* strain GV3101 (pSoup) colonies containing *AcC3H2*-YFP, *AcC3H4*-YFP, *AcC3H8*-YFP, *AcC3H9*-YFP, *AcC3H16*-YFP, *AcC3H21*-YFP, *AcC3H25*-YFP, *AcC3H28*-YFP, *AcC3H34*-YFP, and *AcC3H40*-YFP were grown in 10 mL of liquid LB medium at 28 °C overnight. The culture was then centrifuged and resuspended in buffer (10 mM MgCl<sub>2</sub>, 100 μM acetosyringone, 10 mM MES) to reach an OD<sub>600</sub> of 0.05, and then injected into tobacco (*Nicotiana benthamiana*) leaves. Fluorescence was observed under a confocal laser scanning microscope (Zeiss LSM 710 NLO, Germany). The excitation/emission wavelengths were 488/507 nm for YFP.

#### Plant material and transcriptomic analysis

Pineapple (*Ananas comosus* cv. 'Comte de Paris') fruits were harvested at 70% commercial maturity from a commercial plantation in Xuwen County, Zhanjiang City, and transported to Guangdong Ocean University. Uniform-sized, disease-free fruits were used as the experimental material. The severity of translucency in the flesh was assessed subjectively on a longitudinally cut half of the fruit, with evaluations based on the proportion of the affected area (T0: 0%; T1: 25%; T2: 50%; T3: 75%; T4:100%) [24]. The development stages of fruit translucency were shown in Supplementary Figure S1.

Pulp tissue samples of pineapple were used for RNA-seq analysis following the methods described previously [25]. Transcriptomic analyses were conducted to investigate the expression profiles of the *CCCH* gene family during fruit translucency. The transcript abundances of *AcCCCH* genes were calculated as fragments per kilobase of transcript per million fragments mapped reads (FPKM) and a heatmap based on log<sub>2</sub> (FPKM + 0.01) values was generated using the pheatmap package in R.

#### RNA extraction and quantitative RT-PCR analysis of selected *AcCCCHs*

Total RNA extraction, removal of DNA contamination, cDNA generation of pineapple fruits and quantitative reverse transcription-PCR (qRT-PCR) were performed according to methods previously described [25]. Briefly, qRT-PCR was conducted using the TransStart Tip Green qPCR SuperMix kit (TransGen, China) on a Bio-Rad CFX Duet real-time quantitative PCR instrument (Foster City, CA, USA). The qRT-PCR program was as follows: 95 °C for 1 min, followed by 40 cycles of 95 °C for 10 s and 60 °C for 30 s, and a final step of 95 °C for 30 s. The primer sequences of the selected *AcCCCHs* were listed in Supplementary table S2. The pineapple *Actin* gene (gene ID: Aco003177) was used as the internal reference. The relative gene expression levels for each sample were calculated using the  $2^{-\Delta\Delta CT}$  method. For each analysis, three technical replicates and three biological replicates were performed.

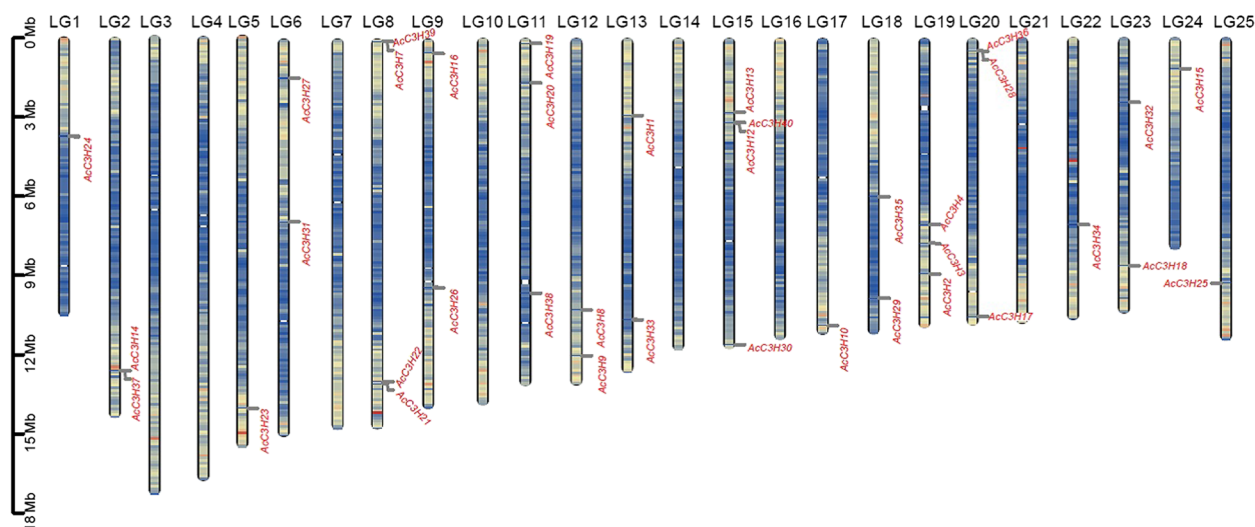
#### Statistical analysis

All statistical analyses were conducted in GraphPad Prism 8.0. Statistical significance between multiple samples were determined by one-way ANOVA with Tukey's test.

## Results

#### Identification of the *CCCH* zinc finger gene family in the pineapple genome

A total of 40 *CCCH* zinc finger genes family members were identified by BlastP search in the pineapple genome database and designated *AcC3H1* to *AcC3H40* (Supplementary Table 1). The proteins encoded by the *AcCCCH* gene exhibited a variety of lengths, ranging from 103 aa (*AcC3H40*) to 2169 aa (*AcC3H2*), with an average length of 591 amino acids. The minimum molecular weight was determined to be 11,207.86 Da (*AcC3H40*), while the maximum molecular weight reached 178,597.74 Da (*AcC3H2*). The predicted isoelectric points (pI) of *AcCCCH* proteins spanned from 4.84 (*AcC3H2*) to 11.65 (*AcC3H29*) (Supplementary Table 1). We found that the pineapple *CCCHs* were distributed on 25 chromosomes (Fig. 1). Chromosomes 1, 5, 17, 22, 24 and 25 contained only one gene, chromosomes 2, 6, 9, 12, 13, 18 and 23



**Fig. 1** Chromosomal distribution of *CCCH* genes family in pineapple. Schematic representations for the chromosomal distribution of pineapple *AcCCCH* genes. The chromosomal number is represented at the top of each chromosome. Gene names were annotated at the outside of the lines

contained two genes, chromosomes 11, 19 and 20 contained three genes, while chromosome 8 and 15 contained four genes.

### Phylogenetic relationships of CCCH gene family

To investigate the evolutionary relationships of CCCH gene family members, we performed a CLUSTALW alignment of 175 CCCH protein sequences, encompassing 40 from *A. comosus*, 68 from *A. thaliana* and 67 from *O. sativa*, and constructed a phylogenetic tree using the ML method of MEGA 7. We showed that AcCCCH proteins were classified into nine distinct categories, designated as Groups I to IX (Fig. 2). The distribution of AcCCCH proteins included all identified categories. Notable variability in the abundance of CCCH zinc finger genes was observed among the various clades, with each clade displaying a gene count ranging from 1 to 9. Group I exhibited the largest numbers of the AcCCCH zinc finger protein family members, followed by the Group II. While Group IV, and VII comprised the fewest AcCCCH proteins, thereby underscoring the diversity and evolutionary divergence of the CCCH zinc finger protein family across these plant species.

### Gene structure, conserved motif and domain analyses of *AcCCCHs*

Structural analysis of the 40 *AcCCCH* zinc finger genes showed that the majority of these genes contain both introns and exons, with only a limited number of exceptions that are devoid of introns. We found that there is significant variability in gene size and the quantity of exons among these genes, with exon counts ranging from 1 to 13 (Fig. 3A and B). Among them, 6 had one exon (15%), 7 had two exons (17.5%), 7 had three exons

(17.5%), and 20 had four or more exons (50%), indicating a complex evolutionary process for the CCCH gene family in pineapple. A total of ten conserved motifs (named motif 1 to motif 10) were found among the 40 AcCCCH zinc finger proteins (Fig. 3C). Subsequently, we found that all *AcCCCH* encoded proteins contained the conserved CCCH zinc finger domain through SMART and Pfam analysis (Fig. 4). Genes that are classified within the same phylogenetic subgroups typically display analogous gene structures and conserved motifs, indicating a potential functional homology among these evolutionary lineages.

### Cis-regulatory element analyses of *AcCCCHs*

To investigate the transcriptional regulatory mechanisms of *AcCCCH* genes, we analyzed potential *cis*-regulatory elements (CREs) in the putative promoter regions of the genes. Various *cis*-elements, including hormone response, plant growth and development, and environmental stress, were found within the *AcCCCH* genes (Fig. 5). We observed that CREs responsiveness to hormones such as GA, SA, ABA, IAA and MeJA. CREs associated with plant growth and development, including light responsiveness, meristem expression, and cell cycle regulation, were also predicted. Furthermore, *cis*-elements associated with environmental stress responses were identified, including low-temperature responsiveness, defense and stress responses, circadian regulation, and anaerobic induction. These results suggest that the differences in the distribution and combinations of *cis*-regulatory elements within the promoters of various *AcCCCH* zinc finger gene family members contributed to their diverse roles in the growth, development, and stress responses of pineapple.





**Fig. 2** Phylogenetic tree of the CCCH zinc finger gene family among different plants. The phylogenetic tree was constructed for the CCCH protein sequences in *Arabidopsis thaliana*, *Oryza sativa* and *Ananas comosus* with the ML (maximum likelihood) method. CCCH genes were divided into nine groups (Group I–XI) according to the clades and bootstrap values. Different colors represented 10 subfamilies

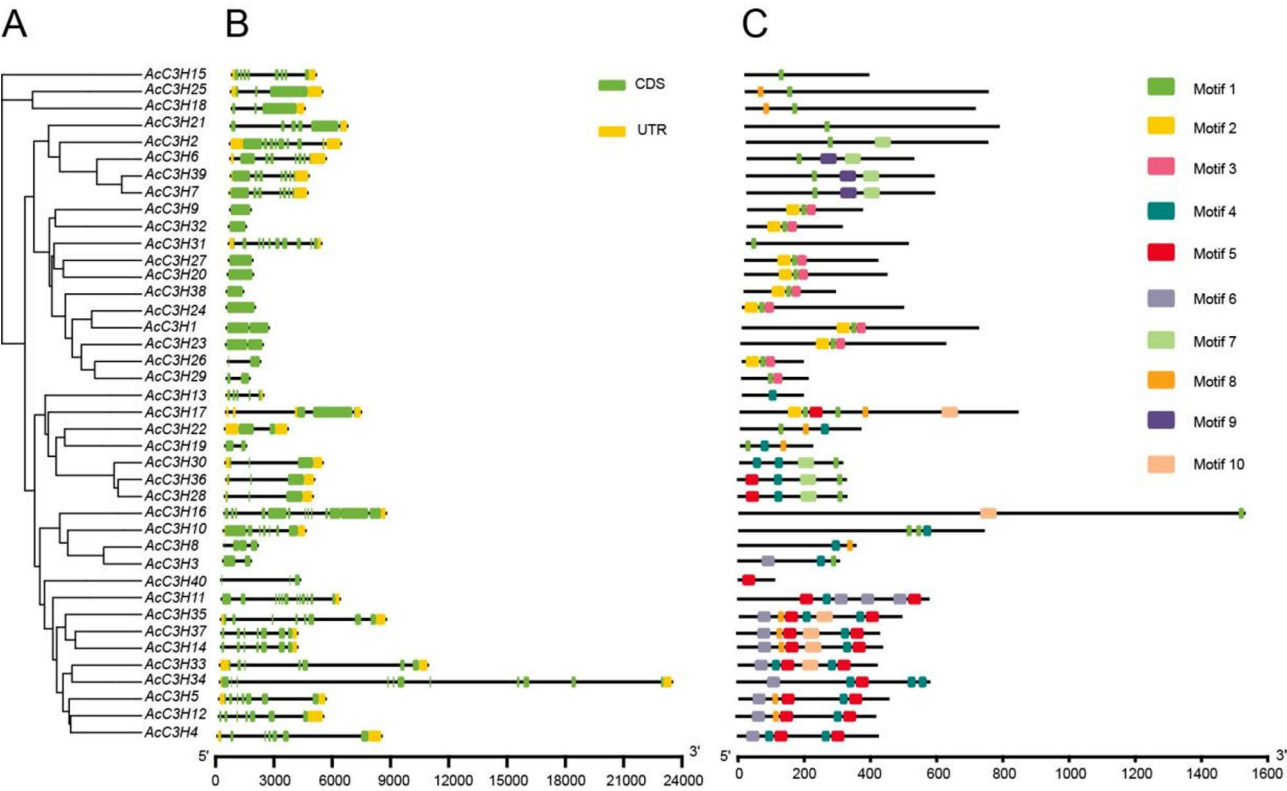
### Expression analysis of *AcCCCH* genes

To explore the spatial pattern of expression of *AcCCCH* genes in different tissues and developmental stages of pineapple, we analyzed the transcriptomic data from previously published research [26]. The various developmental stages of sepals (Se1, Se2 and Se3), petals (Pe1, Pe2 and Pe3), stamens (St1–St6), Gynoecium (Gy 1–Gy 7), ovules (Ov S1–S7) and Fruits (Fr 1–Fr 7) were utilized to investigate the expression patterns of 40 *AcCCCH* genes. Hierarchical clustering analysis revealed distinct temporal regulation patterns among the 40 *AcCCCH* family members (Fig. 6). We found that eight *AcCCCH* genes (*AcC3H3*, *AcC3H8*, *AcC3H11*, *AcC3H16*, *AcC3H19*, *AcC3H27*, *AcC3H32*, and *AcC3H35*) exhibited specific high expression in floral organs, suggesting their potential functional importance during fruit set. During the pineapple fruit development, the expression of *AcC3H10*, *AcC3H14* and *AcC3H31* displayed peak expression at the Fr7 stage, whereas *AcC3H2* and *AcC3H5* showed a progressive decline from the Fr1 stage to Fr7 stage. The

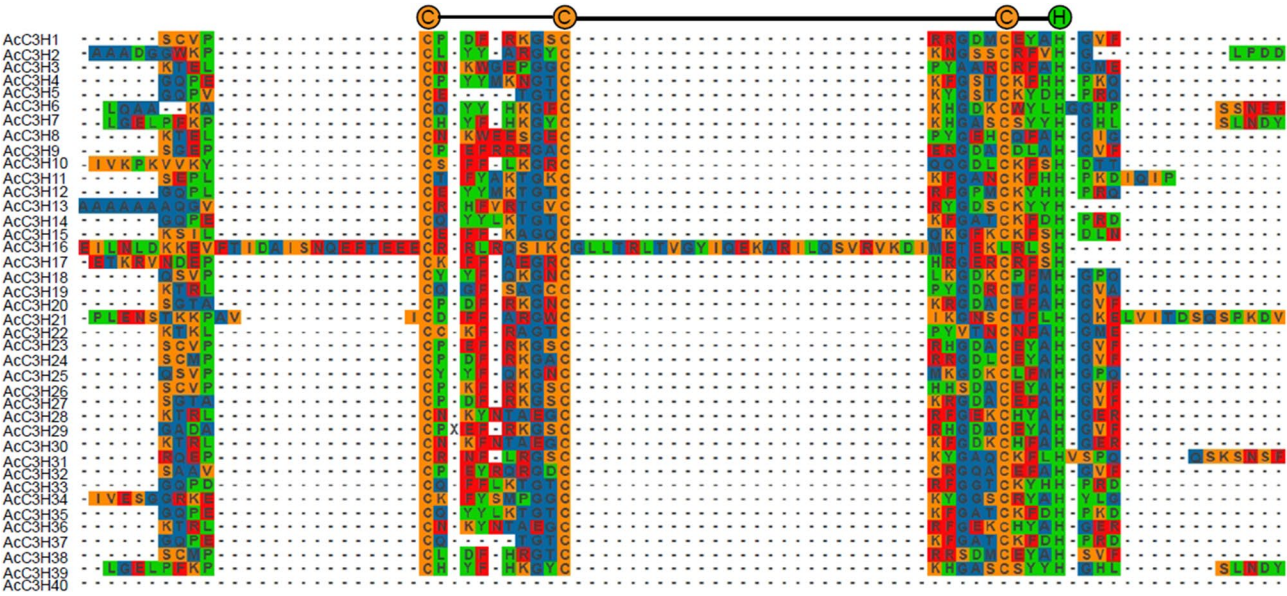
remaining *AcCCCH* family members maintained low to moderate expression levels throughout fruit developmental stages. These stage-specific expression patterns suggest that *AcC3H10*, *AcC3H14* and *AcC3H31* played essential roles in the development of pineapple fruit.

To gain insight into the potential roles of *AcCCCH* genes in pineapple fruit translucency disorder (PFTD), we performed transcriptome profiling on pulp tissues across five developmental stages (T0–T4) via RNA-sequencing. As shown in Fig. 7, specifically, twelve genes (*AcC3H7/9/11/15–18/21/25/30/31/34*) exhibited significant upregulation during late-phase of fruit translucency (Fig. 7). Conversely, four genes (*AcC3H1/8/14/37*) showed progressive downregulation during late-phase of fruit translucency (Fig. 7). The remaining *AcCCCH* gene family members showed a consistent pattern throughout translucency development.

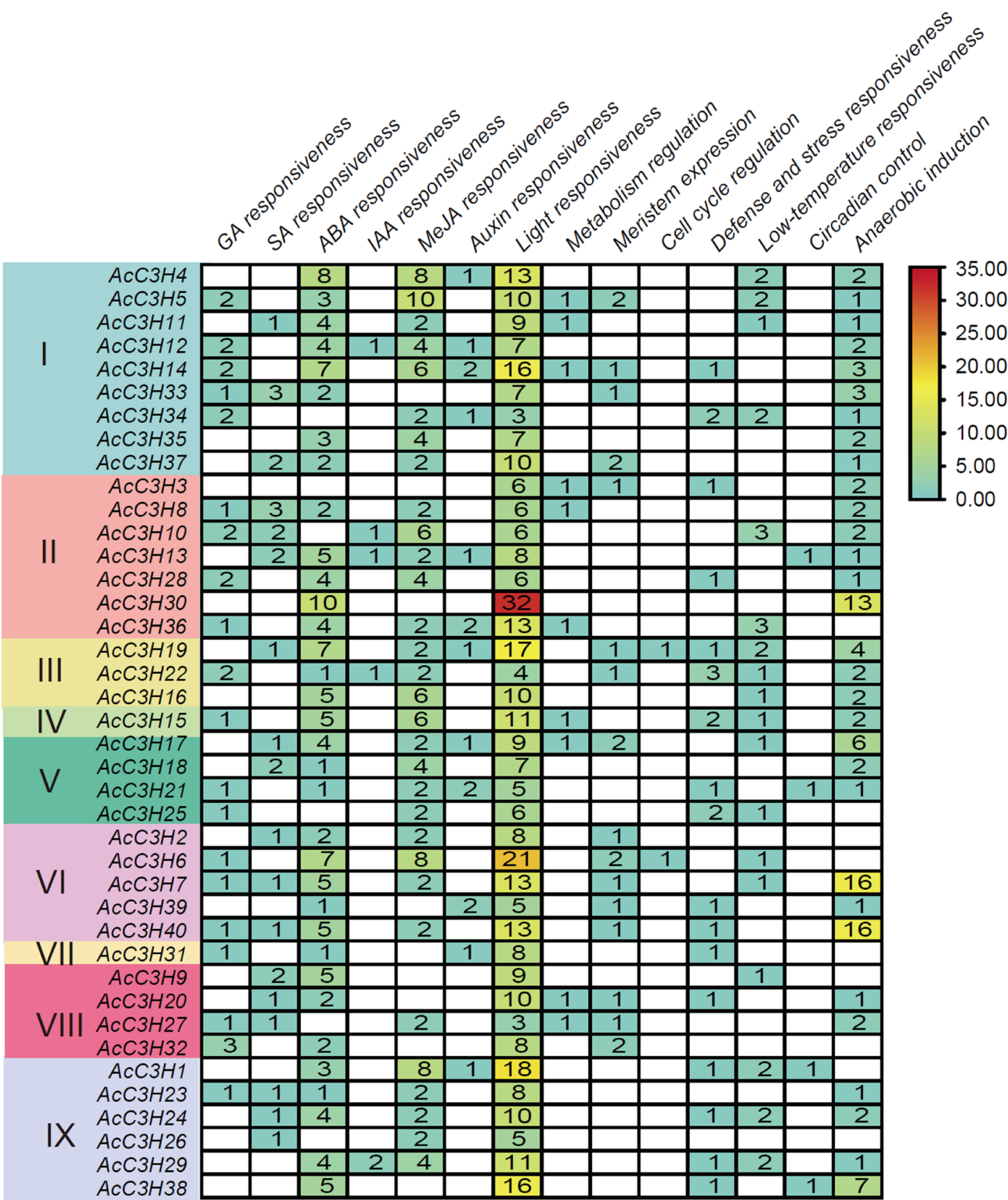
To validate the reliability of our RNA-seq data, we performed qRT-PCR analysis on 12 representative *AcCCCH* genes across five fruit translucency stages (T0–T4). We



**Fig. 3** Phylogenetic relationship, gene structure and protein structure analysis of *AcCCC3H* genes. **A** The neighbor-joining tree was constructed using aligned full-length amino acid sequences. The proteins are named according to their gene name with the CCCH zinc finger number of each protein. **B** Gene structure: The gene structure is presented by black exon (s) and spaces between the black boxes correspond to introns. The sizes of exons and introns can be estimated using the horizontal lines. **C** Schematic representation of conserved motifs in *AcCCC3H* proteins. The length of the motif can be estimated using the scale at the bottom. aa, amino acids



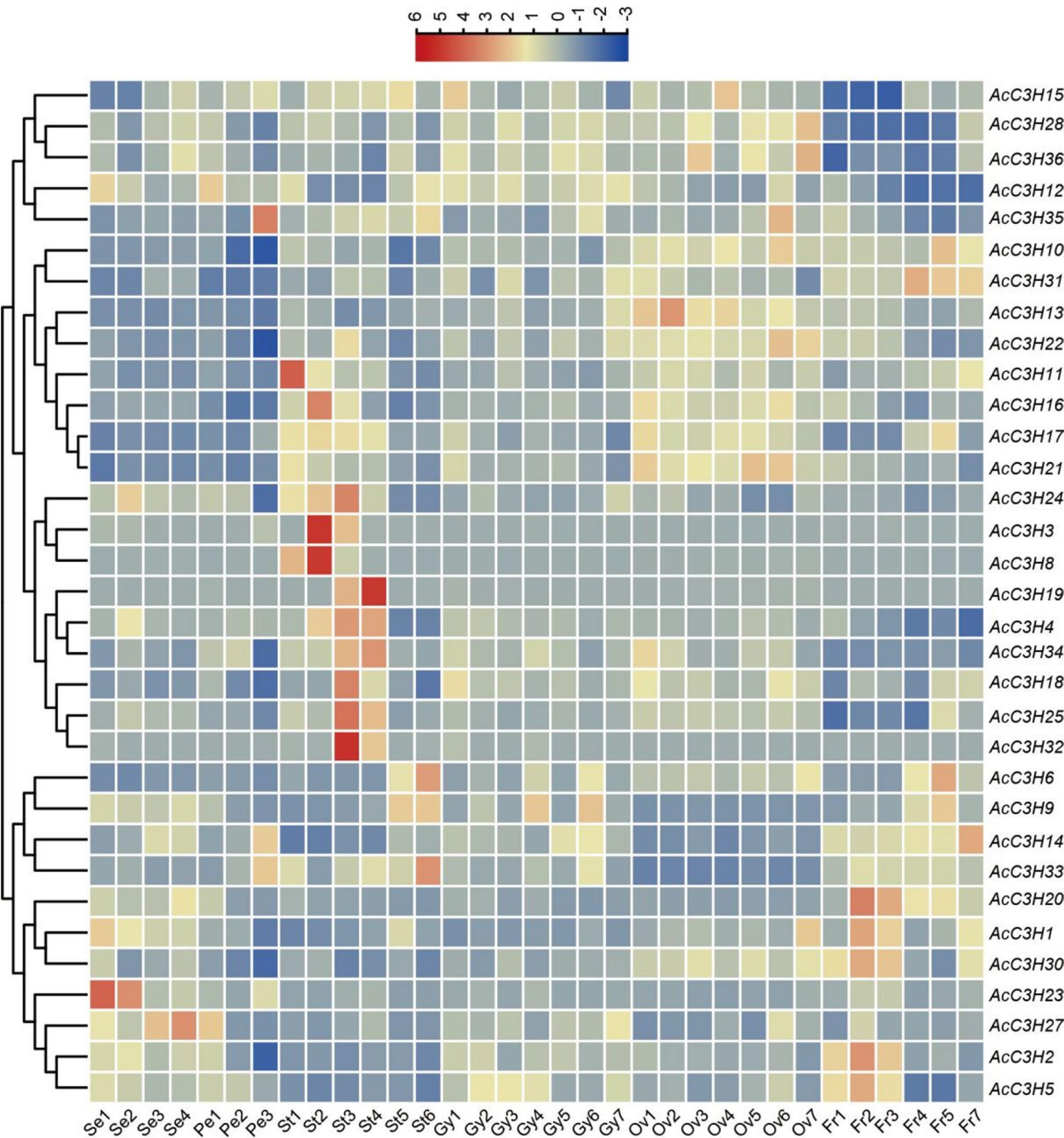
**Fig. 4** Multiple sequence alignment of the *AcCCC3H* proteins. The three cysteine residues and one histidine residue putatively responsible for the formation of the zinc-finger structure are highlighted



**Fig. 5** Distribution of cis-elements on the promoter region of CCCH gene family in pineapple. The cis-element number of each gene is indicated by box

found that *AcC3H7*, *AcC3H18*, *AcC3H25*, *AcC3H31*, and *AcC3H34* exhibited consistent upregulation from the early to late stages (Fig. 8B, E, G, I and K). *AcC3H15*, *AcC3H17*, *AcC3H21* and *AcC3H30*, showed significant induction starting at the mid-stage (Fig. 8E, F, H and J), while *AcC3H1* and *AcC3H14* displayed transient upregulation peaking at T1 followed by a decline (Fig. 8A and D). In contrast, *AcC3H8* expression progressively





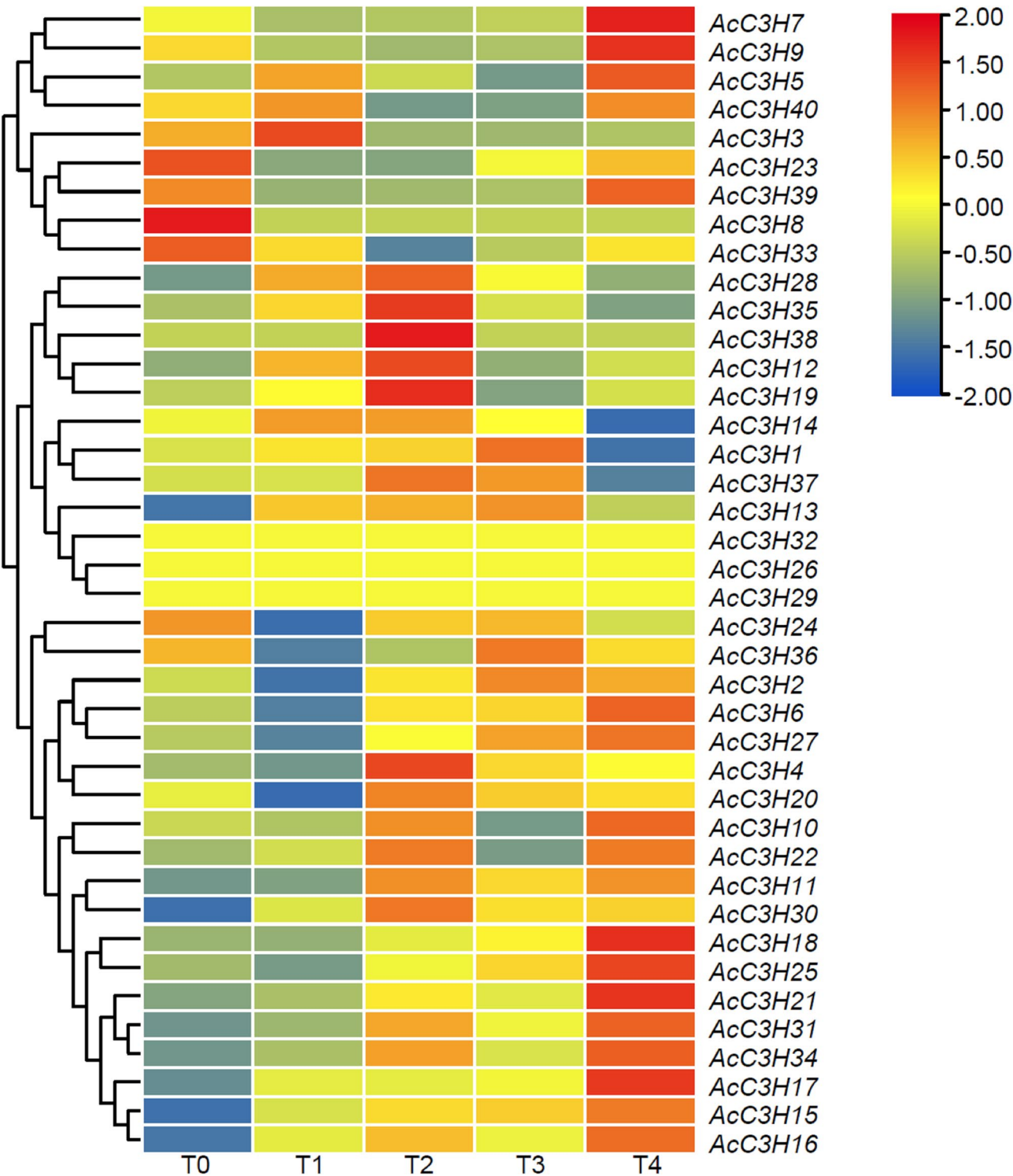
**Fig. 6** Heatmap represents the expression profiles for *AcCCCH* genes in different stages of floral organ formation and fruit development. Expression dynamics based on FPKM (fragments per kilobase of transcript per million fragments mapped). Stamen (St1-St6), Petal (Pe1, Pe2 and Pe3), Sepal (Se1, Se2 and Se3), Gynoecium (Gy1-Gy7), Ovule (Ov1-Ov7), Fruit (Fr1-Fr7)

decreased throughout translucency development (Fig. 8C). Altogether, the qRT-PCR expression profiles of all 12 genes aligned with the RNA-seq trends, confirming the robustness of our transcriptome data and implicating these *AcCCCH* genes in pineapple fruit translucency regulation.

#### Subcellular localization of the selected *AcCCCH* proteins

To determine the subcellular localization of *AcCCCH* proteins, we selected ten *AcCCCH* genes representing distinct phylogenetic groups, including *AcC3H34* and *AcC3H4* (Group I), *AcC3H8* and *AcC3H28* (Group II), *AcC3H16* (Group III), *AcC3H21* and *AcC3H25* (Group V), *AcC3H2* and *AcC3H40* (Group VI), and *AcC3H9* (Group

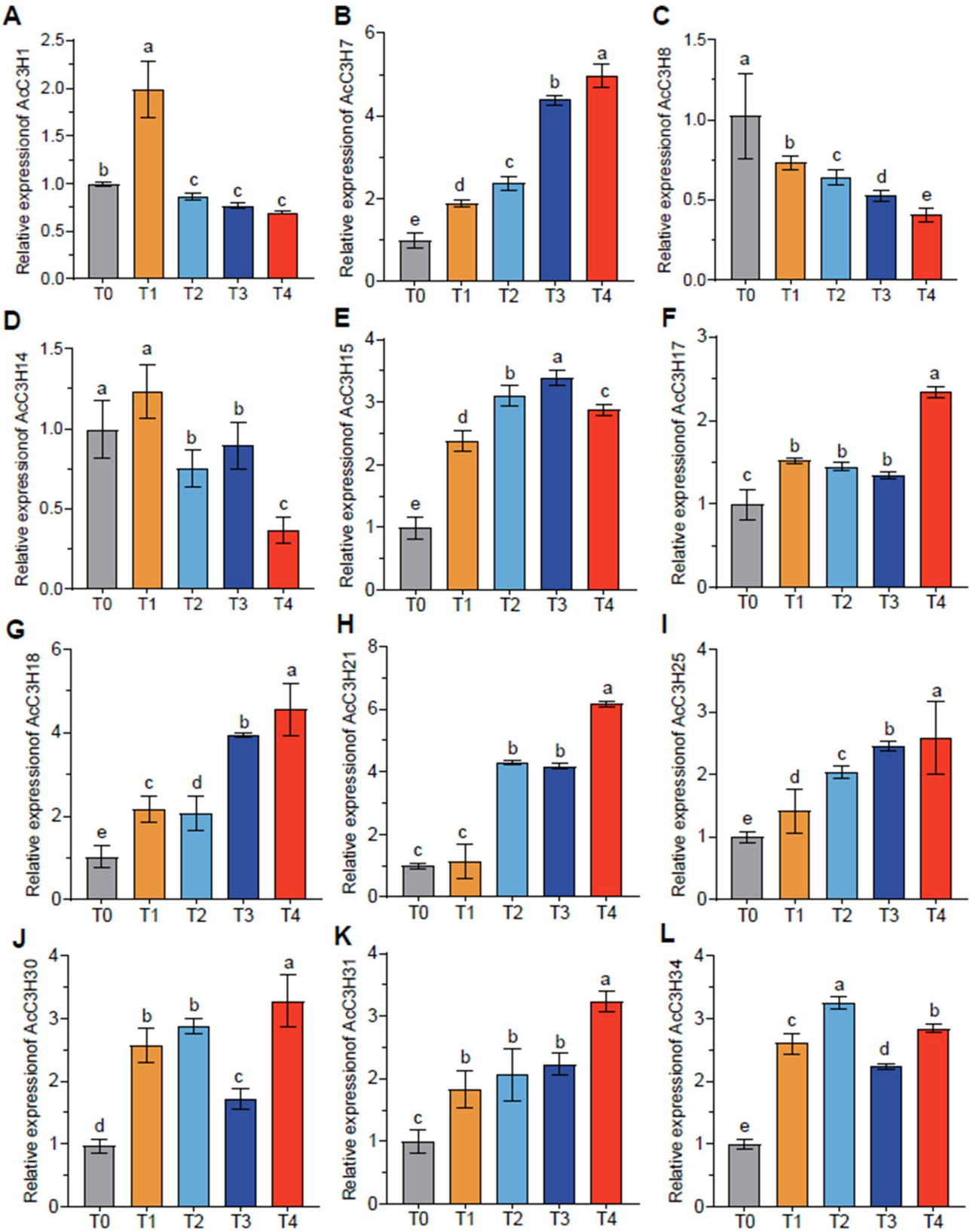




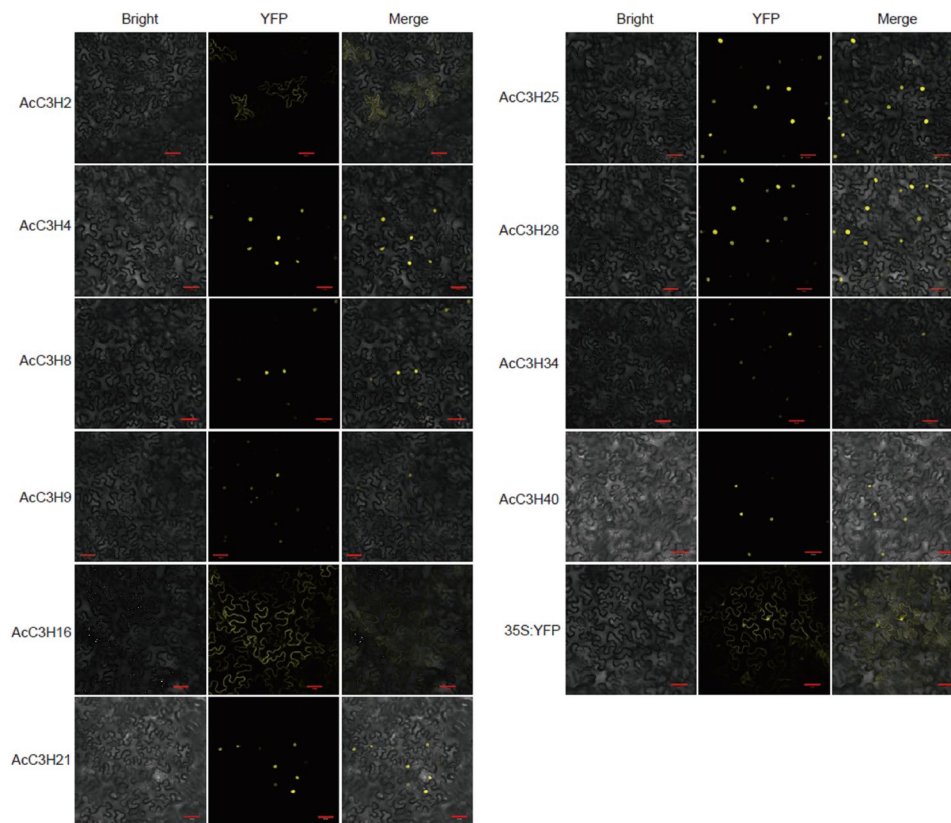
**Fig. 7** Heatmap represents the expression profiles for *AcC3H* genes in different stages of fruit translucency. Expression dynamics based on FPKM (fragments per kilobase of transcript per million fragments mapped). T0, T1, T2, T3 and T4 represented the severity of translucency in the flesh

VIII). Each gene was cloned and fused in-frame with yellow fluorescent protein (YFP) to generate fusion constructs for localization analysis. All of these constructs were used to transiently express in tobacco (*Nicotiana*

*benthamiana*) leaves. Microscopy analysis showed that *AcC3H4*-YFP, *AcC3H8*-YFP, *AcC3H9*-YFP, *AcC3H21*-YFP, *AcC3H25*-YFP, *AcC3H28*-YFP, *AcC3H34*-YFP, and *AcC3H40*-YFP fusion proteins localized exclusively to the



**Fig. 8** qRT-PCR analysis of 12 representative *AcCCCH* genes in different stages of fruit translucency. Representative *AcCCCH* genes included *AcC3H1*, *AcC3H7*, *AcC3H8*, *AcC3H14*, *AcC3H15*, *AcC3H17*, *AcC3H18*, *AcC3H21*, *AcC3H25*, *AcC3H30*, *AcC3H31*, and *AcC3H34*. All the experiments were conducted independently at least three times. Each value represents means  $\pm$  SD (n=3). Different letters indicate significant differences,  $P < 0.05$ , one-way ANOVA together with Tukey's multiple comparison test were used in significance statistical analysis



**Fig. 9** Subcellular localization of pineapple CCCH-ZF proteins. AcCCCHs-YFP fusion proteins were transiently expressed in Tobacco (*Nicotiana benthamiana*) leaves and subcellular localization was observed by confocal laser-scanning microscopy. The merged pictures of the yellow fluorescence channel (right panels) and the corresponding YFP (middle panels), bright field (left panels) are shown. The empty vector pC29-35 S:YFP was used as a control. Bar = 50  $\mu$ m

nucleus, whereas AcC3H2-YFP and AcC3H16-YFP were localized at the plasma membrane (Fig. 9). These results align with the *in silico* predictions of AcCCCH proteins by the online tool (Supplementary Table 1). Overall, the nuclear localization of most AcCCCH protein supports their putative role in transcriptional regulation.

## Discussion

The CCCH-type zinc finger (CCCH-ZF) transcription factor family plays pivotal regulatory roles in plant growth and developmental processes as well as abiotic and biotic stresses [12, 27–29]. The role of CCCH-ZF family genes in drought stress, heat stress, low temperature stress and salt stress has been identified in the tropical crops, such as Pitaya [17], Pepper [30] and Banana [18]. However, there is little knowledge on the roles of the CCCH-ZF family in pineapple, especially the formation of fruit quality. In this study, we identified 40 AcCCCH genes and classified them into nine subfamilies (Fig. 2). Analyzing gene structure and motif composition provides valuable insights into the genetic relationships within multi-gene families. Within each subfamily, AcCCCH genes exhibited consistent gene structures and

motif patterns (Fig. 3), suggesting that genes with similar structures and conserved motifs may share functional similarities. Moreover, we showed that the pineapple CCCH-ZF family genes were clearly classified into different evolutionary clades compared to the CCCH-ZF family genes in *A. thaliana* and *Oryza Sativa* (Fig. 2). The evolutionary relationships among members of the CCCH-ZF family exhibit significant variation across species, indicating that these proteins may fulfill distinct functions in different plants.

Consistent with other species, the promoters of the CCCH-ZF genes in pineapple contain a variety of elements associated with biotic and abiotic stress responses, hormonal regulation, as well as growth and developmental processes (Fig. 5). Numerous investigations in other species demonstrated that plant CCCH genes mediate ABA biosynthesis and signaling, thereby regulating developmental and adaptive processes [31, 32]. In rice, the CCCH protein SWOLLEN ANTHET WALL 1 (SAW1) positively regulates the development of anther walls through the GA signaling pathway [33]. It has been shown that Poplar *PaC3H17* and its target, *PaMYB199*, constitute an auxin-responsive functional complex that



facilitates cambium division through a dual regulatory mechanism [34]. We observed that the promoter region of the *CCCH-ZF* genes mainly included ABA or JA responsive cis-element in pineapple (Fig. 5), suggesting the *CCCH* genes may regulate a variety of biological processes through multiple hormone pathways.

The spatiotemporal expression patterns of *AcCCCH* genes in pineapple, have demonstrated the tissue-specific characteristics of different members (Fig. 6). Specifically, our analysis revealed that eight *AcCCCH* genes (*AcC3H3*, *AcC3H8*, *AcC3H11*, *AcC3H16*, *AcC3H19*, *AcC3H27*, *AcC3H32*, and *AcC3H35*) exhibited significantly elevated expression levels in floral organs (Fig. 6), suggesting that these genes potentially involved in pineapple flowering. It has shown that the specific roles of *CCCH*-type zinc finger transcription factors in fruit ripening and fruit quality. For instance, MaCCCH33-like2/MaEBF1/MaABI5-like complex in banana is crucial for regulation of fruit softening during ripening [15]. *CCCH*-type zinc finger protein, *PbdsZF*, was identified as a key regulator of stone cell formation in pear fruit [35]. We also showed that *AcC3H7*, *AcC3H18*, *AcC3H25*, *AcC3H31*, and *AcC3H34* exhibited sustained upregulation across five fruit translucency stages (Figs. 7 and 8), further suggesting the important role of the *CCCH* zinc finger genes in fruit translucency. Despite these insights, the specific roles of *CCCH*-type zinc finger transcription factors in fruit translucency remain largely unexplored and require further experimental investigation.

The spatial localization of a protein within the cellular environment can provide insights into its functional role [36]. In our investigation, we showed that the pineapple *CCCH* zinc finger proteins displayed distinct localization patterns. Our finding indicated that eight pineapple *CCCH* zinc finger proteins fused with YFP exhibited exclusive nuclear localization (Fig. 9), as observed with *CCCH* zinc finger proteins from *Arabidopsis* [37], rice [38], and poplar [13]. Notably, we found that two *CCCH* proteins, *AcC3H2* and *AcC3H16* located on cytoplasm plasma membrane (Fig. 9), implying that functional differentiation of members of the *CCCH* zinc finger gene family may occur. A larger number of *CCCH* zinc-finger proteins can function as activators or repressors of transcription, and their nuclear localization is consistent with this function [39]. Given that most of *AcCCCH* zinc finger proteins may be regulate growth, development and stress responses at transcriptional levels. However, further experimental investigations need to validate the transcriptional regulation of pineapple *CCCH* genes.

## Conclusion

Altogether, these results shed new light on structural, evolutionary, and subcellular localization of the pineapple *CCCH-ZF* gene family that suggest functional

diversification of these regulatory proteins. Moreover, the expression profiles indicate that *AcCCCH* genes may be involved in fruit translucence. Furthermore, our findings improve the understanding of the pineapple *CCCH* zinc finger gene family and contribute to future biological studies of *AcCCCH* genes in fruit development and fruit translucence.

## Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-06927-7>.

Additional file 1: Figure S1. The developmental stages of fruit translucency. Table S1. Physicochemical properties of *CCCH* zinc finger gene family members in pineapple. Table S2. Primers were used for qRT-PCR analysis and constructing the vectors.

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## Authors' contributions

D.L. designed the experiments and revised the manuscript; Z.Y. and W.J. performed the gene family analysis and collected the sample; W.J. and L.W. performed the qRT-PCR and constructed the vectors; Z.Y. and W.J. wrote the original manuscript; Z.Y. and D.L. reviewed and edited the manuscript.

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## Data availability

The pineapple (*Ananas comosus* L.) genome sequence information was obtained from the Phytozome website (version: *Ananas comosus* v3; [https://phytozome-next.jgi.doe.gov/info/Acomosus\\_v3](https://phytozome-next.jgi.doe.gov/info/Acomosus_v3)). The original RNA-seq data of pineapple floral organs used in this study were obtained from the European Nucleotide Archive (ENA) under accession number PRJEB38680. Data on the different stages of pineapple fruit development were obtained from <http://pineapple.zhangjisenlab.cn/pineapple/html/download.html>. Transcriptome profiling on pulp tissues across five developmental stages were generated from the National Center for Biotechnology Information (NCBI) under accession number PRJNA1235514. The datasets supporting the conclusions of this article are included within the article and its additional files.

## Declarations

### Ethics approval and consent to participate

The experimental research and method on pineapple species comply with relevant institutional, national, and international guidelines.

### Consent for publication

Not applicable.

### Competing interests

The authors declare no competing interests.

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